

Smaller Families and the Fall in Global Absolute Poverty ^{*}

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June 2026

Abstract

The number of people living below the World Bank’s \$3.00/day line fell by about 1.5 billion between 1990 and 2019, a decline conventionally credited to economic growth. Because the line divides household consumption by household size, falling fertility raises measured consumption per person mechanically, even when household resources are unchanged. I measure this channel directly, transporting the children-per-mother distribution across household surveys for 22 developing countries that were home to 85% of the 1990 poor. Across these countries, falling family size lifted about 167 million people over the \$3.00/day line—14% of their combined poverty decline, and from Ghana (7%) to Mexico (44%) country by country. In 11 of the 22 countries measured poverty would have risen, or risen further, had family size not fallen. The same channel is 20% of the decline in child poverty and 18% of the decline among people in households with children. The share is stable across equivalence scales and at the higher \$4.20 and \$8.30 lines, and standard growth/redistribution accounting hides the channel inside “growth.” Census fertility data reaching back to 1960 show that in most surveyed countries the steepest fertility declines predate the consumption-survey window, so the channel’s full contribution extends well before the poverty data begin.

Keywords: Poverty; fertility; family size; equivalence scales

JEL Classification: I32, J13, O15

^{*}Data and code for replication are available at <https://github.com/adamlooney/global-fertility-poverty>. This research received no external funding and the author reports no conflicts of interest.

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1 Introduction

The decline in global absolute poverty since 1990 is among the most-cited facts in development economics. The World Bank measures poverty as household consumption divided by household size, so a household crosses the poverty line when its consumption per person rises—which can happen either because the household earns more or because it has fewer people to support. Over the same period, fertility fell sharply across the developing world. This paper asks how much of the measured poverty decline is the mechanical consequence of smaller families rather than rising household resources.

The channel is an accounting one. When a mother has fewer children, the household’s consumption is divided among fewer people, so measured consumption per capita rises even if total household consumption is unchanged, and some members cross the line. The estimates are not the causal effect of fertility decline on income, labor supply, or human capital—those channels may be large, but they are not the focus here. The focus is the part of the poverty decline that standard per-capita accounting records as growth because the denominator changed. The design therefore requires no instrument for fertility.¹

The microdata are the paper’s evidence. Applying a rank-preserving transport of the children-per-mother distribution to household surveys from 22 countries that were home to 85% of the 1990 poor, I find that falling family size lifted about 167 million people over the \$3.00/day line between 1990 and 2019—14% of those countries’ combined poverty decline. The country shares range from Ghana (7%) to Mexico (44%), largest where growth was slow and smallest where it was rapid. In 11 of the 22 countries, measured poverty would have risen, or risen further, had family size not fallen. Falling family size also accounts for 20% of the decline in child poverty and 18% of the decline among people in households with children.

A single worldwide percentage is not the right summary. The global net decline is a small

¹The twin-birth instruments common in this literature are in any case not quasi-random in the poorest settings, where multiple births co-vary with maternal health (Bhalotra and Clarke, 2019); the absence of an instrument is a feature of an accounting question, not a gap in the design.

residual of large offsetting flows (poverty rose in parts of Africa even as it collapsed in Asia), so a worldwide ratio swings with the poverty line and the window. I therefore lead with the country-level shares and the absolute count of people lifted (Section 4.1).

For children, the effect is not that poor children are “missing” from the count. Decomposing the change shows that the fall in child poverty comes overwhelmingly from fewer children sharing a fixed household budget—the denominator channel, 112% of the net child effect. The composition channel, which changes which children are counted, is small and runs the other way (-12%; the two sum to one hundred percent by construction), because richer mothers reach zero children fastest. The counterfactual is the same real children living in smaller rather than larger families, with more consumption each—not poor children who are never born and so never counted.

The result reframes a standard accounting. Growth/redistribution decompositions in the tradition of Datt and Ravallion (1992) assign about 75% of the decline to growth, but they condition on the per-capita distribution and so record the demographic channel as growth: roughly 18% of measured “growth” is the family-size denominator. The paper draws on, and distinguishes itself from, four literatures: growth/redistribution accounting (Datt and Ravallion, 1992); the sensitivity of poverty comparisons to household economies of scale (Lanjouw and Ravallion, 1995; Jolliffe and Tetteh-Baah, 2024); macro and dependency-ratio accounts (Eastwood and Lipton, 1999; Ravallion, 2016; Wietzke, 2020; Cruz and Ahmed, 2018); and differential-fertility and cohort decompositions (Vogl, 2016; Armentano, Niehaus and Vogl, 2026). These are summarized in Table 7 (Appendix E). The contribution is a within-household transport at the level of the mother, holding household resources fixed and reading exact headcount counterfactuals at near-global coverage; a decomposition of the child effect into resource dilution and selection; and a census extension showing the surveys open only after the steepest fertility declines.

The method imposes one survey wave’s distribution of children per mother onto another wave’s population, within urban/rural cells, holding each household’s consumption and its

non-children fixed, and recomputes the headcount. Working at the level of the mother rather than the household separates falling fertility from changes in how households co-reside, and matches the unit of the companion United States study. Identification rests on rank preservation in the children distribution—defensible because the fertility–income gradient is stable over the period (Section 6)—and the experiment conditions on children, the household members who consume far more than they earn. The genuine uncertainty is how ties are broken among mothers with equal children counts: adversarial welfare orderings put the pooled share between 2% and 23% around the headline, while orderings that mirror the data—a Gaussian copula at the observed welfare–fertility correlation, or mothers’ ages—reproduce it (Section 6). Section 7 closes with what the finding implies for interpreting the past decline, projecting the future one, and designing the statistics themselves.

2 Data and Measurement

The evidence is household surveys for 22 countries, together 85% of the people below \$3.00/day in 1990. From each survey I take the household roster (children aged 0–17, the mother of each child, and other members), the consumption aggregate, an urban/rural indicator, and sampling weights. Children are linked to their mother by a probabilistic rule—among co-resident women of plausible age, preferring the head’s spouse or a female head. Where the survey also records a mother line-number, it validates the rule rather than replacing it: the two agree for 95% of children (median across the surveys that record one). The quantity the surveys measure is therefore co-resident children under 18 per co-resident mother—shaped by births, but also by child mortality, children leaving home, and fostering—which is why the paper speaks of family size rather than fertility. Appendix B describes each country’s surveys, welfare aggregates, and access routes. Every wave is anchored to the official World Bank headcount: I rescale survey consumption so the share below \$3.00/day (2021 PPP)

reproduces the Poverty and Inequality Platform figure (World Bank, 2024).² Anchoring makes levels match the published series by construction and absorbs level differences from deflators, purchasing-power adjustments, and consumption aggregates across countries and years; it is exact in practice, with the anchored headcount reproducing the official figure to within 0.12 percentage points in every one of the 87 country-waves (Table 3). What matters for the decomposition is not the level but the shape of the distribution near the line, and that is what the transport moves, not what the anchoring pins.³

Two facts about the surveys discipline the design. First, the consumption surveys begin late relative to the fertility transition: in most countries the first usable wave is in the 1990s or later, after fertility had already fallen for a generation. Second, the surveys do not cover the whole developing world. To place the survey-based estimates in their longer and wider context I use the children distributions recorded in IPUMS-International censuses (Minnesota Population Center, 2024), which carry no welfare but record each household’s children and their mothers directly, for 77 countries back to 1960. Section 5 uses these censuses to show that the surveyed window captures only part of the transition, and to extend the decomposition backward where a census predates the first survey.

The headline window is 1990 to 2019, chained over decadal sub-periods; per-country results run to the latest available wave. I report poverty at \$3.00/day in the main text and at the higher \$4.20 and \$8.30 lines (2021 PPP) as robustness.

3 The Family-Size Transport

Consider two survey waves of country c , an earlier t_0 and a later t_1 . I ask what the wave- t_0 poverty rate would have been if mothers had had the wave- t_1 distribution of children, holding

²The main line is the World Bank’s \$3.00/day line in 2021 PPPs, adopted in the June 2025 update and replacing the \$2.15 line in 2017 PPPs; the increase reflects higher national poverty lines among low-income countries more than price change. I use the PIP vintage retrieved 2026-06-10. Because every wave is anchored to the official headcount, level revisions across PIP vintages are absorbed by the anchoring.

³Two survey comparability breaks are worth naming: India’s HCES 2022–23 redesign relative to NSS 2011–12, and Nigeria’s sampling-frame change between 2018–19 and 2022–23. The chains cross these breaks at the same waves PIP’s own series does, and each wave is anchored to its own PIP headcount.

everything else fixed. Within urban/rural cells, I rank mothers by their number of children and match each to the corresponding rank of the target wave’s distribution; I then replace each mother’s children with this counterfactual count, sum to the household, keep its total consumption and its non-children unchanged, recompute consumption per capita, and recount the poor at \$3.00/day. The difference from the actual headcount is the contribution of family-size change. I run the experiment in both directions and average, and I chain decadal sub-periods rather than performing a single multi-decade transport.

The key assumption is rank preservation: a mother high in the children distribution in one wave is treated as high in the counterfactual. This is credible only if the fertility–income gradient is stable; Section 6 shows that it is, and that forward and backward transports agree once sub-periods are chained. The experiment conditions on children throughout. Children consume far more than they earn, so holding household consumption fixed while varying their number is the appropriate thought experiment; adults earn, so they are held fixed and the co-residence margin is left out by construction. I report the effect on three populations: all persons, children (persons under 18), and all persons living in households with children.

Formally, let household h have total consumption C_h , A_h non-children, and a set of resident mothers each with K_m children, so $K_h = \sum_{m \in h} K_m$. A mother here is a woman with at least one co-resident child under 18; women without co-resident children are not units of the transport, so movements across the childless boundary are excluded by construction and examined at the household level in Appendix D. Per-capita consumption is $x_h = C_h/(A_h + K_h)$ and the line is z . The transport assigns each mother a counterfactual child count and sums them to a household total K_h^{cf} (children not linked to a co-resident mother are held fixed), so $x_h^{\text{cf}} = C_h/(A_h + K_h^{\text{cf}})$. The three headcounts are

$$\begin{aligned} H^{\text{all}} &= \frac{\sum_h w_h (A_h + K_h) \mathbf{1}\{x_h < z\}}{\sum_h w_h (A_h + K_h)}, & H^{\text{child}} &= \frac{\sum_h w_h K_h \mathbf{1}\{x_h < z\}}{\sum_h w_h K_h}, \\ H^{\text{kidsHH}} &= \frac{\sum_h w_h (A_h + K_h) \mathbf{1}\{K_h > 0\} \mathbf{1}\{x_h < z\}}{\sum_h w_h (A_h + K_h) \mathbf{1}\{K_h > 0\}}, \end{aligned}$$

where w_h is the survey weight. The counterfactual headcounts replace x_h with x_h^{cf} , and K_h with K_h^{cf} in the weights of H^{child} . The family-size effect on each population is the observed minus the counterfactual headcount, reported both as a rate and, multiplied by the relevant population base, as a count of people. The all-person effect is the symmetric average of the forward ($t_0 \rightarrow t_1$) and backward transports; the child and with-children effects are reported for the forward direction, where the counterfactual child population is well defined.

4 Results

4.1 The distribution of the effect across countries

Figure 1 and Table 1 report the country estimates. The family-size share of the all-person poverty decline ranges from Ghana (7%) to Mexico (44%), and the ordering is systematic: the share rises as the pace of growth falls. Where growth was rapid (Indonesia, China, Ghana), income gains swamped the denominator and the share sits at the bottom of the range; in the slow, late-stage declines (Mexico, Brazil), family size did much of the work. The reason is mechanical. The family-size margin varies far less across countries than the pace of the total decline, so the ratio is ordered mostly by its denominator: where growth was fast the total decline is large and the share small; where growth was slow, the same demographic arithmetic is a large fraction of a small total.

The lower tail of the growth distribution turns the same arithmetic into a starker statement. In 11 of the 22 countries, the family-size margin exceeded the entire net change in measured poverty; without it, poverty would have risen. In 7 of them, poverty rose outright: Nigeria, Côte d’Ivoire, Kenya, Malawi, Egypt, Mali, and Niger. In these countries falling family size was not a contributor to decline; it was the main measured force holding poverty down while incomes stagnated or deteriorated. The Sahel windows are short—their comparable surveys begin only in 2018—so their margins are small; the long-window cases, Côte d’Ivoire, Kenya, and Malawi, carry the point. Peru makes it across a crisis and recovery:

its headcount ends the period close to where it began even as the family-size margin pushed steadily down throughout. For these countries a share is undefined or misleading, so Table 4 reports the margins in percentage points. Family size is thus not a marginal correction but a first-order component of the decline where poverty fell, and the principal stabilizer where it did not.

The absolute counts summarize the same evidence without a fragile denominator. The number of people below \$3.00/day fell from 2.3 billion in 1990 to 837 million in 2019, a decline of 1463 million. The 22 surveyed countries—85% of the 1990 poor—deliver the family-size contribution directly: smaller families lifted about 167 million people over the line between 1990 and 2019, 14% of their combined decline.⁴ The aggregate chains decadal sub-periods and credits each country only within its surveyed window, so it omits family-size effects in the years the surveys do not cover—China’s large post-2002 decline above all—and is conservative for that reason. At the higher lines the corresponding counts are 140 million at \$4.20 and 49 million at \$8.30; what is stable across lines is the share of each line’s decline (Table 6), not the count, since far fewer people crossed the \$8.30 line at all.

The global net decline is a residual: poverty collapsed across Asia while it rose across much of Sub-Saharan Africa, where fertility was the main force holding it down. Dividing the family-size count by that small residual gives a ratio unstable enough that at the \$8.30 line, where the global headcount did not fall at all, it is undefined. The country-level shares and the absolute counts are the right summaries, and I do not report a single global percentage.

For comparison, each country’s actual \$3.00/day headcount over the same period is shown in Appendix A (Figure 5); the family-size channel of Figure 1 is the part of that observed decline the transport attributes to smaller families.

⁴Pooled shares move by a point or two with the weighting: 14% people-weighted across the surveyed countries, 13% for the median country, and 15% counts-weighted over the pooled micro sample (the Table 1 pooled row). Nor does any single country carry the number: dropping any one country moves the people-weighted share by at most 2 percentage points.

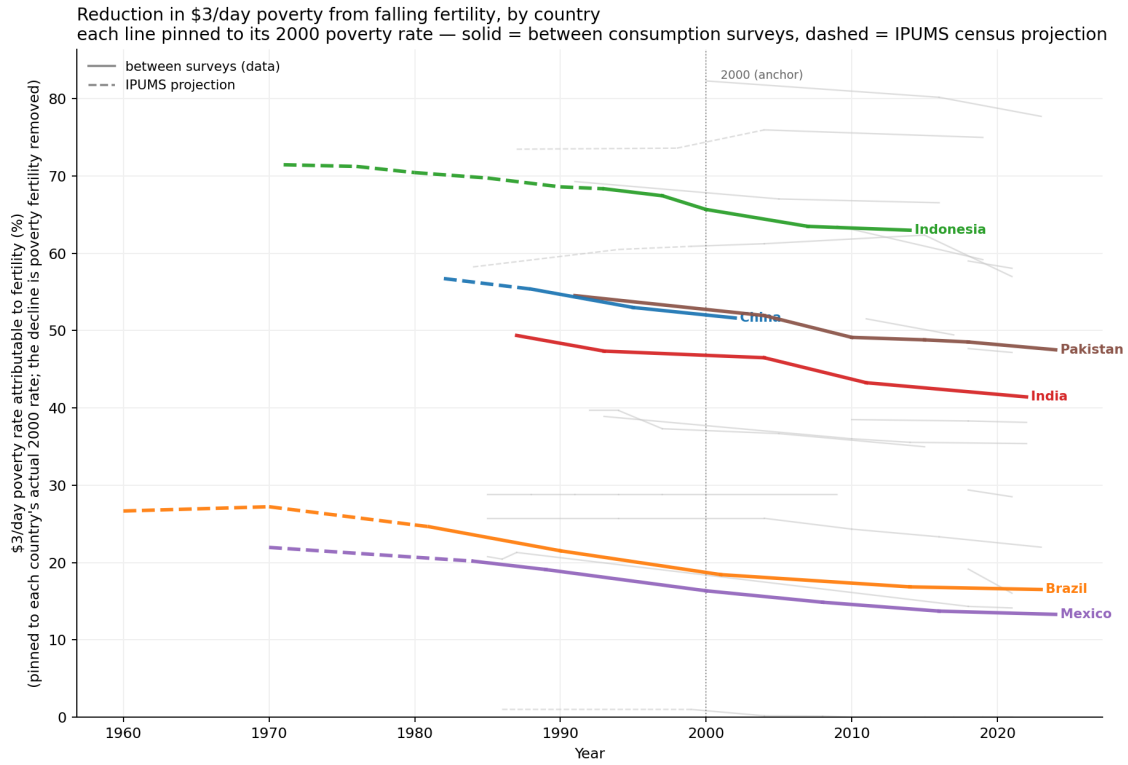


Figure 1: The fall in \$3.00/day poverty attributable to falling family size, by country. Each line is anchored to the country's actual poverty rate in 2000 and traces the poverty path implied by the family-size channel alone; solid segments span the survey window, dashed segments are projected back before the first survey using census fertility (Section 5).

Table 1: Family size and the poverty decline, by country

Country	Window	Total decline (pp)	Family-size share of the decline		
			All (%)	Child (%)	With kids (%)
Mexico	1989–2024	13	44	54	51
Brazil	1990–2023	27	19	24	24
India	1987–2022	47	17	21	20
Ethiopia	1999–2021	27	14	14	13
South Africa	1993–2022	27	13	27	22
Pakistan	1991–2024	56	13	12	12
China	1988–2002	35	11	16	13
Rwanda	2000–2023	44	10	9	9
Indonesia	1993–2014	52	10	11	10
Ghana	1991–2016	41	7	13	11
Pooled (micro)	pooled	—	15	20	18

Notes: Rank-preserving transport of the children-per-mother distribution, chained over sub-periods, anchored to official PIP \$3.00/day headcounts. “All” is the all-person symmetric share; “Child” and “With kids” are the forward-leg shares of child poverty and of poverty among persons in households with children. Clean-decline countries (decline > 10 points) only; country windows differ (column 2), so the pooled row pools each country’s own window. The pooled row weights by counts of people over the pooled micro sample; the people-weighted 1990–2019 aggregate in the text (14%) weights countries by their contribution to the covered decline.

4.2 Child poverty: denominator, not composition

For children, the family-size effect could in principle work through two channels: a denominator channel, in which a child in a smaller family has more consumption per person, and a composition channel, in which differential fertility changes which children are counted. Decomposing the two shows the effect is almost entirely the denominator: it accounts for 112% of the net child effect, with the composition channel making up the small remainder (-12%; the two sum to one hundred percent by construction). The composition channel is adverse on net, a sign implied by rank preservation: when the children distribution contracts, it is the high-consumption mothers who move to zero children first (the gradient is negative), so the children who remain are, if anything, more concentrated among the poor, not less. In Mexico the childless share of the top consumption quintile rises from 45% to 81% over the period, against 7% to 20% in the bottom quintile. The family-size denominator effect for children is therefore more resources per surviving child, not a statistical artifact in which poor children are simply no longer born and counted. A household-level decomposition shows the effect runs almost entirely through the intensive margin—fewer second, third, and higher-order children rather than the move across the childless boundary (Appendix D).

4.3 Comparison with growth/redistribution accounting

The standard tool for decomposing poverty change is the growth/redistribution split of Datt and Ravallion (1992), which attributes a fall in poverty either to a rising mean of the per-capita distribution (“growth”) or to a change in its shape (“redistribution”). Applied to the same anchored surveys, it assigns 75% of the decline to growth and 25% to redistribution—the familiar finding that growth dominates. But this accounting takes the per-capita distribution as its primitive, and a fall in family size raises the per-capita mean mechanically. Figure 2 shows that roughly 18% of what the decomposition labels “growth” is in fact the family-size denominator. The two ratios are mutually consistent: 75% of the decline labeled growth, of which 18% is family size, recovers the pooled family-size share of

the decline. The transport is what separates households becoming richer from households becoming smaller; the growth/redistribution split cannot.

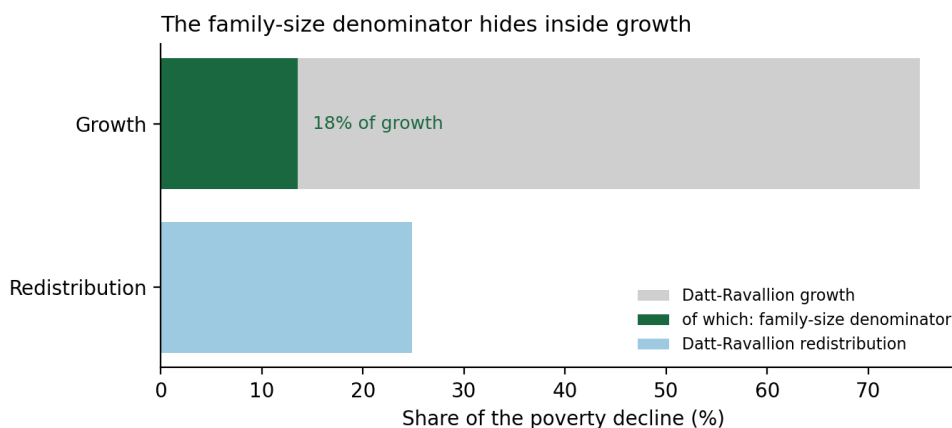


Figure 2: The family-size denominator hidden inside the Datt–Ravallion growth term: a share equal to 18% of measured “growth” is the shrinking family-size denominator, not households becoming richer.

5 The transition before the data: a longer horizon

The surveyed window understates fertility’s role for a simple reason: in most countries the consumption surveys begin after the fertility transition is well under way. Census records, which reach back to 1960 for 77 countries, make this visible: Figure 3 plots completed fertility (children ever born to women aged 40–49), which had been falling for decades before the first consumption survey.

Where a census predates a country’s first survey, I extend the decomposition backward, imposing the census children-per-mother distribution on the earliest survey wave exactly as the transport does between waves. The extension reaches 1971 in Indonesia, 1960 in Brazil, 1970 in Mexico, and 1982 in China (Figure 1, dashed segments; Appendix C). Where official poverty statistics exist before the first survey, the extension yields a clean pre-survey share of the decline: 17% in Indonesia over 1985–1993 and 9% in China over 1982–1988, of the same order as those countries’ post-1990 survey estimates. The channel was already at work before

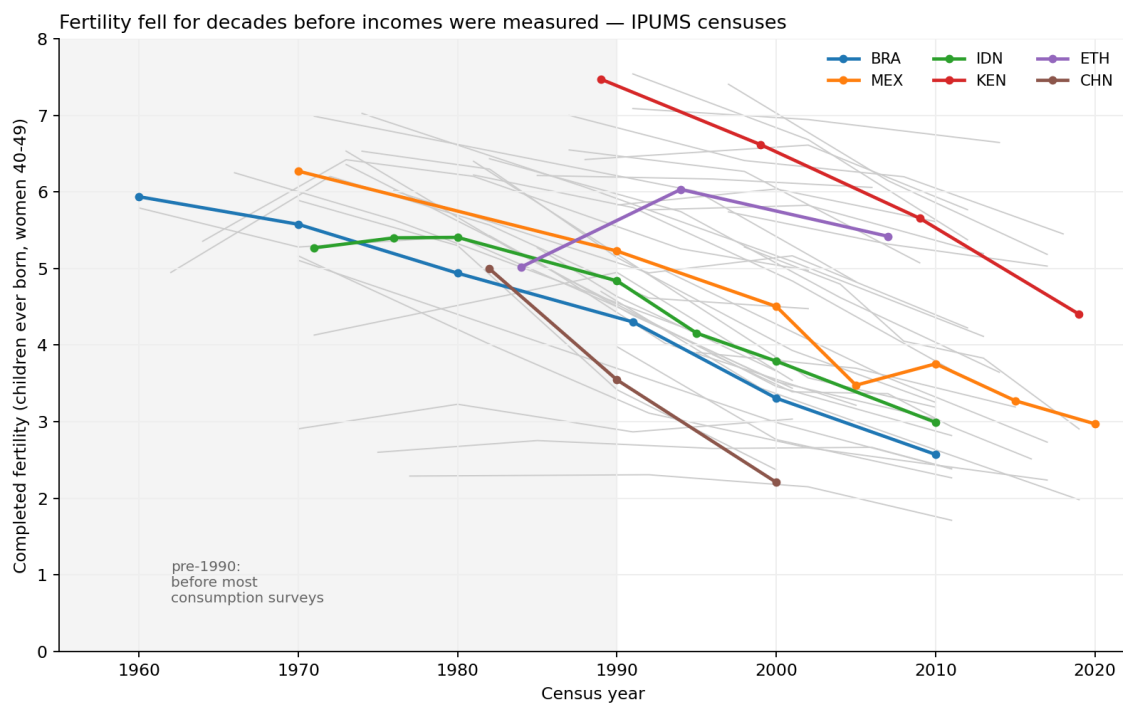


Figure 3: Completed fertility (children ever born to women aged 40–49) from IPUMS-International censuses, 77 countries. Fertility fell steeply for decades before consumption surveys began, so the survey-window estimates omit the steepest part of the transition.

the surveys begin, so the post-1990 estimates omit an earlier phase of the same denominator effect.

6 Robustness and identification

Two concerns bear on the estimates: the measurement convention and the identifying assumption. The convention matters little. Because the transport holds each household’s measured welfare aggregate fixed and moves only the children distribution, the share does not depend on whether a survey reports income or consumption—each transport leg uses a single wave’s welfare, so no leg mixes concepts—and it is nearly invariant to the equivalence scale and the poverty line: moving one mother’s children changes household size by little, so the per-capita metric and sub-proportional scales, and the \$4.20 and \$8.30 lines, give almost the same pooled share (Tables 5 and 6; Appendix D).

The identifying assumption is rank preservation: each mother keeps her position in the children distribution across waves. The diagnostic most sensitive to a violation is the agreement of the forward and backward transports, since re-ranking would break their symmetry. Chaining decadal sub-periods cuts the median forward–backward relative gap from 49% over a single multi-decade transport to 30%, and nearly eliminates it where it was largest (India, from 84% to 7%). The residual gap is the poverty line sweeping the local density as the distribution shifts, not rank churn—which is why the estimates chain sub-periods. Rank preservation fails outright only if the fertility–income gradient reverses (Vogl, 2016), and it does not: the poorest quintile holds more children per mother than the richest in every wave of every country (22 of 22 countries), the gap compressing but never crossing zero (Figure 4), and the within-wave slope of children on the consumption rank is negative throughout (a non-negative slope appears in 0 of 80 waves; the least steep is -0.38).

The unidentified part of the estimate is how ties are broken among mothers with equal children counts. The random-tie headline of Table 1 sits between the welfare-ordered ex-

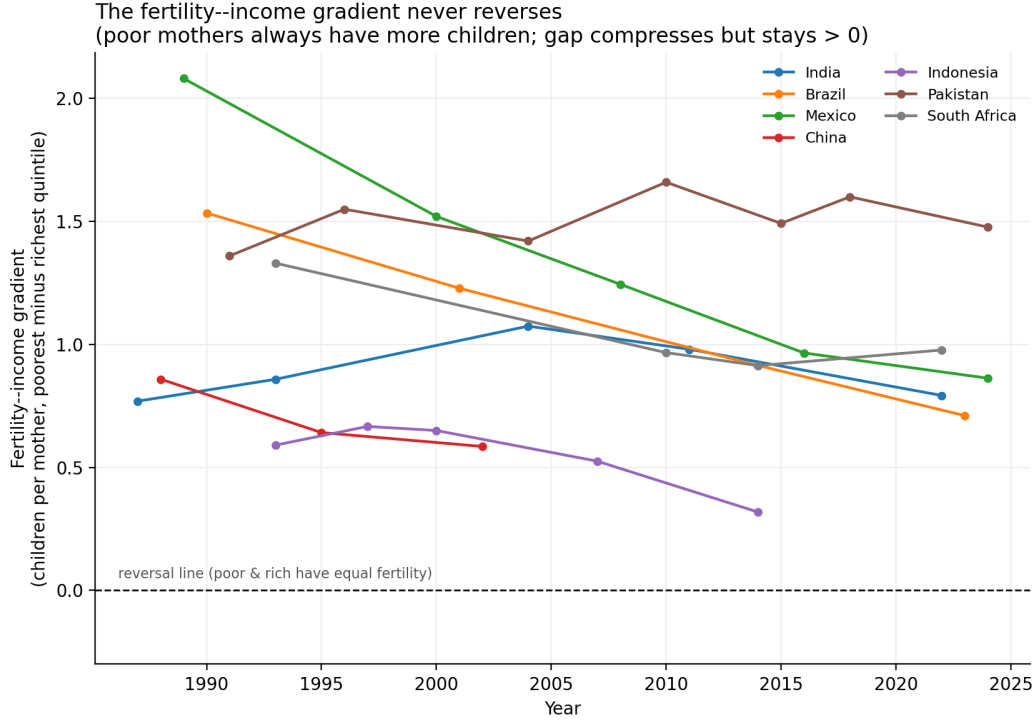


Figure 4: The fertility–income gradient over time: the poorest consumption quintile always has more children per mother than the richest; the gap compresses but never reverses.

tremes that assign counterfactual children rich-first and poor-first; pooled over the clean-decline countries those extremes give a wide band of 2%–23% (Table 2), and the assumption-free bounds that drop rank preservation entirely span zero. The band is wide because, for the large low-income countries, many households cluster at the same children counts, so the ordering within ties does a lot of work. The random tie-break is the right point estimate within that band: conditional on the children count, nothing in the data favors assigning the counterfactual to richer or to poorer mothers, so the rich- and poor-first orderings are deliberately adversarial bounds, not competing estimates. Two orderings that use the data rather than the adversary bear this out. A Gaussian-copula tie-break, which assigns counterfactual counts within ties at each survey’s own welfare–fertility correlation—the natural reading of “the relationship between fertility and income holds at its typical strength”—gives a pooled 14%; a demographic ordering that gives the higher counterfactual counts to younger mothers, whose fertility is least complete, gives 14% (Table 2). Both sit on the random headline. The

quantitative content of the assumption-free bounds spanning zero is that rank preservation, supported by the gradient diagnostics above, is what carries the identification. Appendix D reports the full set of checks.

Table 2: Family-size share under alternative rank tie-breaking

Country	Rich-first (%)	Copula (%)	Random (%)	Age (%)	Poor-first (%)
Mexico	13	47	44	45	96
Brazil	6	20	19	20	43
India	-1	17	17	18	26
China	4	11	11	10	11
Indonesia	-0	10	10	10	8
Pakistan	-1	13	13	13	22
South Africa	-1	14	13	14	24

Notes: Chained family-size share of the all-person poverty decline under tie-breaking rules for mothers with equal children counts. Rich-first assigns counterfactual children to the richest equal-kids mothers (minimizing the effect), poor-first to the poorest (maximizing it); random (seeded) is the headline. Copula assigns within ties by a Gaussian copula at the survey’s own welfare–fertility correlation; Age gives the higher counterfactual counts to younger mothers, whose fertility is least complete. Discrete headcounts can place the random estimate marginally outside the rich/poor-first range, and where many mothers tie at the same children count the rich-first ordering can flip the sign of the margin; the pooled tie-breaking band is 2%–23%.

7 Implications

The estimates are an accounting statement: they do not require fertility decline to be exogenous or causal for household income, only that the per-capita line divides by household size. Four implications follow.

First, the channel is one-time. A country completes its fertility transition once, and the transition is now essentially complete in Latin America and East Asia, where fertility sits at or near replacement: the tailwind that delivered 14% of the surveyed countries’ poverty decline is largely spent there, and future reductions must come mainly from household resources. The unspent balance sits where poverty remains: countries where children under 15 are still at least 40 percent of the population held 57% of the world’s remaining \$3.00/day

poor in 2019, and Sub-Saharan Africa—an overlapping but distinct cut—held 60%.⁵ As those transitions complete, the same denominator arithmetic that ran through three decades of measured decline will run through Africa’s—and then through no one’s.

Second, where incomes stagnated, falling fertility has hidden it. In the 11 cushion countries of Section 4.1, the statistics flatter income performance: a flat headcount reads as holding ground when household resources were deteriorating and smaller families absorbed the difference. The welfare reading follows. A decline in poverty from rising income, holding family size fixed, is an unambiguous improvement in living standards. A decline from smaller families, holding income fixed, is a real change: parents chose smaller families, and each member commands more consumption. But the per-capita metric books the entire re-division as if it were new resources: it values a child only as a claim on the household budget, and it counts nothing of what parents sought in having children. Revealed preference cuts the other way: where fertility fell by choice, households moved along their own preferred path, and the measured gain need not overstate the welfare gain; where it fell by mandate, as in China, that reading fails too. The paper does not adjudicate. What the accounting establishes is narrower and sufficient: the two kinds of decline are not welfare-equivalent, and the headline statistic cannot tell them apart. Among the clean declines, the family-size component runs from Ghana (7%) to Mexico (44%), so two countries can report similar poverty progress with very different income stories underneath. And because richer mothers reached zero children first, children make up a growing share of the remaining poor under this mechanism.

Third, growth–poverty elasticities calibrated on this period embed the denominator. Estimates of how much poverty reduction a unit of growth delivers (Ravallion, 2001; Bourguignon, 2003; Lakner et al., 2022) are fit to survey means whose growth partly reflects shrinking households: of what standard decompositions record as growth, roughly 18% was

⁵Countries are classified by the 2019 under-15 population share (UN World Population Prospects 2024), a simple marker of how far the fertility transition has run; poor counts are PIP lined-up 2019 estimates. The share is a majority at either cutoff: 68% at a 35 percent threshold, 57% at 40 percent.

the family-size denominator rather than rising household resources (Section 4.3). The embedded component is also unusually potent against poverty because its incidence is pro-poor—poor mothers had the most children, so the same demographic change lifts more people over the line per unit of mean growth than income growth does. An elasticity carried forward from 1990–2019 therefore overstates what income growth alone will deliver where transitions are complete, and understates the total decline still to come where fertility is falling; projections that instead apply distribution-neutral growth to a fixed survey distribution inherit only the second error, by holding household structure fixed while it continues to change.

Fourth, the statistics could say so directly. Demography reports age-standardized mortality rates precisely because raw rates conflate incidence with population composition; poverty measurement faces the same problem and admits the same solution. A demographically standardized series—the headcount recomputed holding the children-per-mother distribution fixed at a base year—would separate the resources margin from the needs margin in the world’s most-watched welfare statistic, and the transport used here is the machinery such a series needs. It would face the same base-year and tie-breaking choices this paper documents, and its magnitudes would carry the same identification band (Section 6); the direction of the correction would not. Published alongside the headline, it would make visible in real time what this paper reconstructs after the fact: how much of measured progress against poverty reflects households becoming richer, and how much reflects families becoming smaller.

Data Availability Statement

All code that constructs the PIP series, ingests and harmonizes each survey, implements the transport and the Datt–Ravallion decomposition, builds the census extension, and generates every table, figure, and number in this paper is available at <https://github.com/adamlooney/global-fertility-poverty>. The analysis draws on publicly released or licensed household-survey microdata that the author is not permitted to redistribute; the

replication repository documents each source and how to obtain it, together with a harmonized variable dictionary. International poverty statistics are from the World Bank Poverty and Inequality Platform (World Bank, 2024) (vintage retrieved 2026-06-10); census children-per-mother margins are from IPUMS-International (Minnesota Population Center, 2024).

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Appendix

The Global Fertility Dividend

A Survey coverage and anchoring

Table 3 lists, for every country and wave, the survey, the year, the welfare concept (consumption or income, read from PIP metadata), the official PIP headcount used for anchoring, the number of households, children per household, and the weighted survey population as a ratio to the UN World Population Prospects (the full table, with the rescale factor and person and child counts, is in `out/coverage_table.csv`). Each wave's per-capita welfare is rescaled so its person-weighted headcount at \$3.00/day reproduces the official PIP figure; the population ratio flags surveys whose weights are relative (self-weighting, ratio near zero) or whose frame is off by more than 15%.

Table 3: Survey coverage and anchoring, by country and wave

Country	Year	Survey	W	PIP H	n_{hh}	Kids/hh	Pop/WPP
Brazil	1981	PNAD / PNADC	Y	0.307	102,219	2.06	0.96
	1990	PNAD / PNADC	Y	0.303	71,558	1.69	0.93
	2001	PNAD / PNADC	Y	0.175	101,423	1.26	0.95
	2014	PNAD / PNADC	Y	0.048	115,021	0.82	0.97
	2023	PNAD / PNADC	Y	0.038	139,014	0.68	1.00
Burkina Faso	2018	EHCVM	C	0.477	7,010	3.46	0.99
	2021	EHCVM	C	0.421	7,176	3.57	0.98
Côte d'Ivoire	1985	CILSS / EHCVM	C	0.136	1,588	3.50	self-wt
	1986	CILSS / EHCVM	C	0.102	1,600	3.38	self-wt
	1987	CILSS / EHCVM	C	0.160	1,600	3.41	self-wt
	2018	CILSS / EHCVM	C	0.217	12,992	2.17	0.93
	2021	CILSS / EHCVM	C	0.208	12,965	2.21	1.01
China	1988	CHIP	Y	0.811	18,997	1.50	0.97
	1995	CHIP	C	0.626	14,929	1.07	1.00
	2002	CHIP	C	0.478	16,035	0.81	0.99
Egypt	1999	HIECS	C	0.000	23,975	2.07	self-wt
	2004	HIECS	C	0.041	47,095	1.76	self-wt
	2008	HIECS	C	0.037	23,428	1.85	self-wt
	2010	HIECS	C	0.015	7,719	1.71	self-wt
	2012	HIECS	C	0.013	7,528	1.71	self-wt
	2015	HIECS	C	0.012	11,988	1.67	self-wt
	2017	HIECS	C	0.031	12,485	1.61	self-wt
	2019	HIECS	C	0.020	11,085	1.62	self-wt
	2021	HIECS	C	0.014	12,463	1.59	self-wt

continued...

Country	Year	Survey	W	PIP H	n_{hh}	Kids/hh	Pop/WPP
Ethiopia	1999	HICES / ESS	C	0.659	16,672	2.64	0.84 [†]
	2004	HICES / ESS	C	0.413	21,595	2.60	0.85 [†]
	2015	HICES / ESS	C	0.330	4,717	2.45	0.88
	2021	HICES / ESS	C	0.386	4,959	2.07	0.80 [†]
Ghana	1991	GLSS	C	0.804	4,523	2.40	self-wt
	2005	GLSS	C	0.609	8,687	1.93	1.01
	2016	GLSS	C	0.390	14,009	1.75	0.97
India	1987	NSS / HCES	C	0.518	127,310	2.22	0.89
	1993	NSS / HCES	C	0.475	113,131	2.01	0.84 [†]
	2004	NSS / HCES	C	0.464	124,644	1.88	0.86
	2011	NSS / HCES	C	0.271	101,662	1.58	0.88
	2022	NSS / HCES	C	0.052	261,746	1.28	0.85
Indonesia	1993	IFLS	C	0.780	7,214	1.91	self-wt
	1997	IFLS	C	0.681	7,371	1.72	self-wt
	2000	IFLS	C	0.657	10,259	1.52	self-wt
	2007	IFLS	C	0.411	12,977	1.28	self-wt
	2014	IFLS	C	0.257	15,143	1.27	self-wt
Kenya	1992	WMS / KIHBS	C	0.320	8,517	2.55	0.95
	1994	WMS / KIHBS	C	0.320	10,857	2.91	1.02
	1997	WMS / KIHBS	C	0.326	10,755	2.38	0.89
	2005	WMS / KIHBS	C	0.446	13,158	2.53	0.99
	2015	WMS / KIHBS	C	0.377	21,773	1.92	0.96
Malawi	2004	IHS	C	0.744	11,280	2.36	0.99
	2019	IHS	C	0.754	11,434	2.27	0.95
Mali	2018	EHCVM	C	0.294	6,602	3.86	0.96
	2021	EHCVM	C	0.361	6,143	3.88	0.96
Mexico	1984	ENIGH (incl. tradicional)	Y	0.149	4,735	2.53	1.04
	1989	ENIGH (incl. tradicional)	Y	0.148	11,531	2.30	0.97
	2000	ENIGH (incl. tradicional)	Y	0.163	10,108	1.66	1.00
	2008	ENIGH (incl. tradicional)	Y	0.101	29,468	1.45	1.01
	2016	ENIGH (incl. tradicional)	Y	0.044	70,311	1.21	0.99
	2024	ENIGH (incl. tradicional)	Y	0.017	91,414	0.93	1.00
Niger	2018	EHCVM	C	0.590	6,024	3.54	0.98
	2021	EHCVM	C	0.605	6,622	3.43	0.98
Nigeria	2010	GHS-P / NLSS	C	0.385	4,839	2.79	0.98
	2018	GHS-P / NLSS	C	0.342	22,110	2.50	1.03
	2022	GHS-P / NLSS	C	0.418	21,916	2.41	1.00
Pakistan	1991	PIHS / HIES	C	0.795	4,794	3.80	self-wt
	1996	PIHS / HIES	C	0.453	14,901	3.27	1.01
	2004	PIHS / HIES	C	0.409	14,695	3.35	0.76 [†]
	2010	PIHS / HIES	C	0.267	16,341	2.92	0.65 [†]
	2015	PIHS / HIES	C	0.169	24,238	2.91	0.75 [†]

continued...

Country	Year	Survey	W	PIP H	n_{hh}	Kids/hh	Pop/WPP
Peru	2018	PIHS / HIES	C	0.165	24,809	2.85	0.92
	2024	PIHS / HIES	C	0.230	30,123	2.67	0.97
	1985	ENNIV/ENAH0	C	0.062	4,994	2.52	self-wt
	1994	ENNIV/ENAH0	C	0.257	3,623	2.26	self-wt
	2004	ENNIV/ENAH0	Y	0.235	19,502	1.67	0.99
	2010	ENNIV/ENAH0	Y	0.112	21,496	1.39	1.02
	2016	ENNIV/ENAH0	Y	0.075	35,785	1.17	1.03
Rwanda	2023	ENNIV/ENAH0	Y	0.059	33,886	0.99	1.01
	2000	EICV	C	0.823	6,420	2.69	0.97
	2016	EICV	C	0.638	14,580	2.10	1.00
Senegal	2023	EICV	C	0.386	15,054	1.83	0.97
	2018	EHCVM	C	0.192	7,156	4.46	1.00
	2021	EHCVM	C	0.179	7,120	3.95	1.01
South Africa	1993	PSLSD / IES / LCS	C	0.446	8,770	1.87	0.92
	2010	PSLSD / IES / LCS	C	0.278	25,328	1.45	0.96
	2014	PSLSD / IES / LCS	C	0.279	23,380	1.25	1.04
	2022	PSLSD / IES / LCS	C	0.174	19,940	1.13	1.12
Tanzania	2011	HBS	C	0.515	10,186	2.52	0.92
	2017	HBS	C	0.513	9,432	2.31	0.94
Uganda	2009	UNPS	C	0.635	2,497	2.83	0.72 [†]
	2019	UNPS	C	0.598	3,074	2.38	0.89

W: C=consumption, Y=income (PIP metadata). Pop/WPP < 0.1 = self-weighting; [†] frame/vintage off >15%.

The displayed country panel in Table 1 reports the clean-decline countries (those whose \$3.00/day headcount fell by more than ten points), for which a share of the decline is well defined; the pooled row is computed over all 22 micro countries, including those where poverty rose or stalled and the family-size effect is reported in percentage points rather than as a share. Table 4 gives every micro country's chained estimate—the total decline, the family-size margin in percentage points, and, where the decline exceeds ten points, the all-person, child, and with-children shares.

Table 4: Family-size share of the poverty decline, every country

Country	Window	Decline (pp)	Family size (pp)	Share (%)	Child (%)	With kids (%)
Brazil	1990–2023	27	5.0	19	24	24
Burkina Faso	2018–2021	6	0.5	—	—	—
Côte d'Ivoire	1987–2021	-5	7.1	—	—	—
China	1988–2002	35	3.7	11	16	13
Egypt	1999–2021	-1	1.6	—	—	—
Ethiopia	1999–2021	27	3.9	14	14	13
Ghana	1991–2016	41	2.7	7	13	11

continued...

Country	Window	Decline (pp)	Family size (pp)	Share (%)	Child (%)	With kids (%)
India	1987–2022	47	8.0	17	21	20
Indonesia	1993–2014	52	5.4	10	11	10
Kenya	1992–2015	-6	4.7	—	—	—
Malawi	2004–2019	-1	1.0	—	—	—
Mali	2018–2021	-7	0.9	—	—	—
Mexico	1989–2024	13	5.7	44	54	51
Niger	2018–2021	-1	0.9	—	—	—
Nigeria	2010–2022	-3	0.4	—	—	—
Pakistan	1991–2024	56	7.1	13	12	12
Peru	1994–2023	0	3.7	—	—	—
Rwanda	2000–2023	44	4.6	10	9	9
Senegal	2018–2021	1	3.1	—	—	—
South Africa	1993–2022	27	3.5	13	27	22
Tanzania	2011–2017	0	2.1	—	—	—
Uganda	2009–2019	4	4.4	—	—	—

Chained mother-level transport, all 22 micro countries. Share (family-size pp ÷ decline) is shown only where the decline exceeds ten points; elsewhere the margin is reported in pp, as it offsets a flat or rising headcount. “Child” and “With kids” are forward-leg shares.

Figure 5 plots each country’s actual \$3.00/day headcount over the study period; the family-size channel isolated in Figure 1 (main text) is the part of each of these observed declines the transport attributes to smaller families.

B Data sources, country by country

This section records, for each country, the survey series and producing agency, the waves used, the welfare aggregate, and how the data are obtained; sample sizes, headcounts, and children-per-household counts are in Table 3, and the ingest code is in the replication repository. Where a welfare aggregate was built from the survey’s item modules rather than distributed ready-made, that is noted.

Brazil. The National Household Sample Survey (PNAD, 1981, 1990, 2001, and 2014) and its continuous successor (PNADC, 2023), from the Brazilian Institute of Geography and Statistics, publicly downloadable. Welfare is household income, the World Bank’s concept for Brazil. The 1981 and 1990 record layouts were reconstructed from archival documentation.

Burkina Faso, Mali, Niger, and Senegal enter through the harmonized survey of the West African Economic and Monetary Union (EHCVM, 2018 and 2021), from the national statistics institutes via the World Bank Microdata Library. The instrument is standardized—a common consumption aggregate, a common roster, and public-use files—so one ingest serves all four.

China. The Chinese Household Income Project (CHIP, 1988, 1995, and 2002), distributed by ICPSR under license and obtained through the University of Utah. Welfare follows the World Bank’s concept for China: income in 1988, when the official series is income-based,

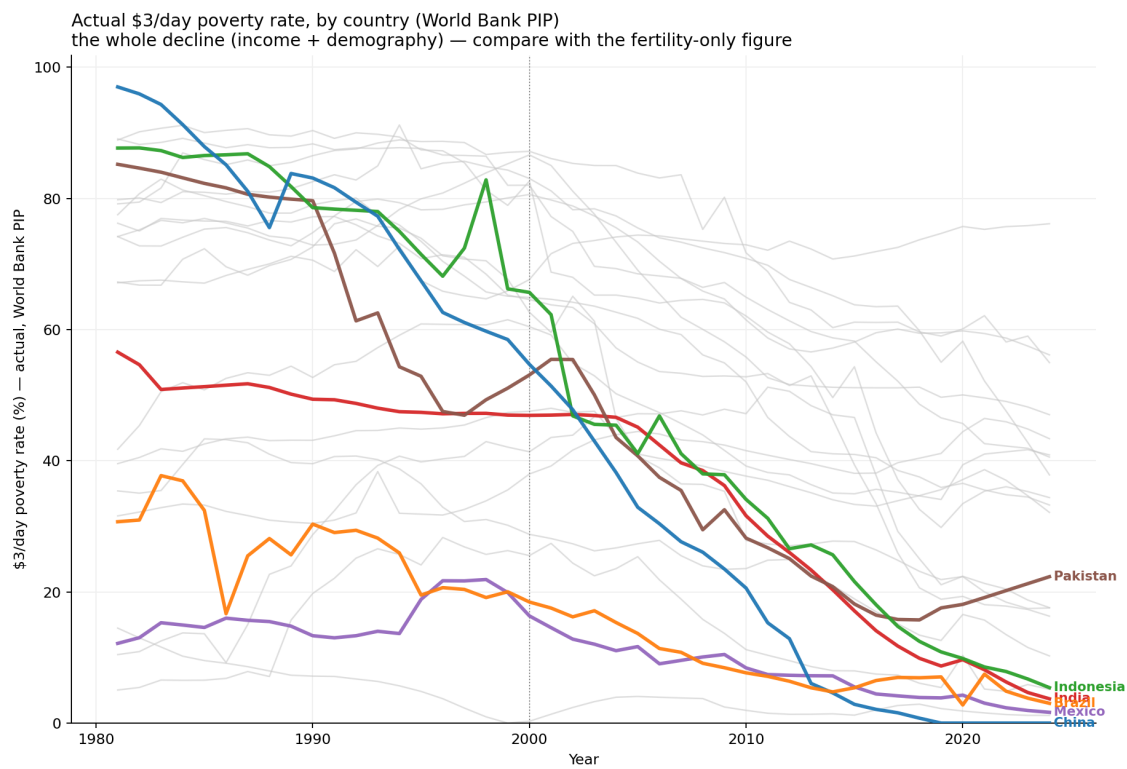


Figure 5: The actual \$3.00/day poverty headcount, by country, over the study period (World Bank Poverty and Inequality Platform).

and consumption thereafter. CHIP releases separate, unweighted rural and urban samples and no household weights, so each is anchored to its own official rural or urban headcount and the two are combined at the official population shares.

Côte d’Ivoire. The Living Standards Surveys (CILSS, 1985–1987) and the harmonized EHCVM (2018 and 2021), from the National Statistics Institute via the World Bank Microdata Library. Welfare is consumption; the CILSS waves ship a constructed aggregate and carry normalized survey weights. The two survey families, three decades apart, are transported as two episodes.

Egypt. The Household Income, Expenditure and Consumption Survey (HIECS, nine waves 1999–2021), from CAPMAS, obtained through the Economic Research Forum under license. Welfare is total household expenditure. The roster records each child’s mother directly, and the recorded link resolves for nearly all children. Measured \$3.00/day poverty in Egypt is low throughout.

Ethiopia. The Household Income, Consumption and Expenditure Survey (HICES, 1999 and 2004) and the Socioeconomic Survey (ESS, an LSMS-ISA panel, 2015 and 2021), from the Central Statistical Agency via the World Bank Microdata Library. Welfare is consumption; the panel waves need a current-resident filter to count only present members.

Ghana. The Living Standards Surveys (GLSS, 1991, 2005, and 2016), from the Ghana Statistical Service. Welfare is consumption; the 1991 wave is the World Bank’s harmonized (SHIP) version, which carries the constructed aggregate and is self-weighting.

India. The National Sample Survey consumption rounds (NSS, 1987, 1993, 2004, and 2011) and the Household Consumption Expenditure Survey (HCES, 2022), from the Ministry of Statistics, retrieved through its NADA portal. Welfare is monthly per-capita consumption; for HCES 2022 it is built from the item-level files with the official recall conversions, reproducing the published aggregate. The 2011-to-2022 comparability change is discussed in Section 2.

Indonesia. The Indonesia Family Life Survey (IFLS, 1993, 1997, 2000, 2007, and 2014), from RAND under a data-use agreement. Welfare is consumption. IFLS covers thirteen provinces and carries no sampling weights, so its waves are anchored to the national headcount; the roster records each child’s mother directly.

Kenya. The Welfare Monitoring Surveys (WMS, 1992, 1994, and 1997) and the Integrated Household Budget Surveys (KIHBS, 2005 and 2015), from the Kenya National Bureau of Statistics. Welfare is consumption; the 1992 aggregate is built from the survey modules. The three WMS rosters use inconsistent membership definitions, so a sensitivity chain drops the 1997 wave.

Malawi. The Integrated Household Surveys (IHS, 2004 and 2019), from the National Statistical Office via the World Bank Microdata Library. Welfare is consumption.

Mexico. The Income and Expenditure Survey (ENIGH, 1984, 1989, 2000, 2008, 2016, and 2024), from the National Institute of Statistics, publicly downloadable, with the early waves from the historical series. Welfare is household income.

Nigeria. The General Household Survey panel (2010) and the Living Standards Surveys (NLSS, 2018 and 2022), from the National Bureau of Statistics via the World Bank Microdata Library and the Bureau’s portal. Welfare is consumption; the 2022 aggregate is built from the survey modules. The sampling frame changed between 2018 and 2022, as noted in Section 2.

Pakistan. The Integrated Household Survey (PIHS, 1991) and the Household Integrated Economic Survey (HIES, 1996, 2004, 2010, 2015, 2018, and 2024), from the Pakistan Bureau of Statistics, publicly downloadable, with the 1991 wave through the World Bank. Welfare is consumption. The 1991 wave is self-weighting, and the 1996 wave carries no design weights, so its households are post-stratified to the known urban share.

Peru. The National Household Survey (ENAH, 2004, 2010, 2016, and 2023), from the National Institute of Statistics, and the earlier Living Standards Surveys (ENNIV, 1985 and 1994), a World Bank Living Standards Measurement Study effort. ENAH is publicly downloadable; the ENNIV waves come through the World Bank Microdata Library. Welfare follows the official series: consumption through 1994, income from 2004. Each wave is anchored to its own headcount, so the concept change falls at the same node the published series does.

Rwanda. The Integrated Household Living Conditions Surveys (EICV, 2000, 2016, and 2023), from the National Institute of Statistics through its NADA portal. Welfare is consumption; the 2000 aggregate is assembled from the survey modules.

South Africa. The Project for Statistics on Living Standards and Development (PSLSD, 1993), the Income and Expenditure Surveys (IES, 2010 and 2022), and the Living Conditions Survey (LCS, 2014), from SALDRU and Statistics South Africa via DataFirst. Welfare is consumption expenditure.

Tanzania. The Household Budget Surveys (HBS, 2011 and 2017), from the National Bureau of Statistics through its NADA portal. Welfare is consumption. The 2000 wave is unusable—its public release omits the consumption diary—so the chain begins in 2011.

Uganda. The National Panel Survey (UNPS, an LSMS-ISA panel, 2009 and 2019), from the Uganda Bureau of Statistics via the World Bank Microdata Library. Welfare is consumption.

Three sources are common to every country. The official poverty statistics used for anchoring and the lined-up 2019 estimates of Section 4.1 are from the World Bank Poverty and Inequality Platform (World Bank, 2024) (vintage 2026-06-10). Population totals and the under-15 shares used in the implications are from the United Nations World Population Prospects. The census children-per-mother distributions behind the long-horizon extension are from IPUMS-International (Minnesota Population Center, 2024). In every survey, children are matched to a mother by the probabilistic rule of Section 3; where a survey also records a mother line-number—Egypt and Indonesia throughout, several others in later waves—that link is the validation sample of Section 2.

C The census extension to the pre-survey period

Section 5 uses census fertility to extend the decomposition before the first consumption survey. The censuses are the IPUMS-International samples (Minnesota Population Center, 2024), which record each household’s children and the mother of each child but carry no welfare. The construction holds a survey wave’s PIP-anchored welfare distribution fixed and replaces its children-per-mother distribution with that of a census from an earlier year, transported the same rank-preserving way as between two survey waves; the difference in the headcount is the family-size contribution over the census-to-survey interval. The construction

imports no assumed income distribution: the welfare is the country's own survey, and only the fertility marginal is the census's.

I treat the extension as reliable for a country only where the census and the survey agree on fertility in their overlapping years—specifically where the census mother-linked children distribution matches the survey's at the anchor wave—so that the backward extrapolation is a genuine continuation of the same gradient rather than a splice across incompatible measurements. Figure 6 shows the extension for the countries that pass this check. Where official poverty statistics exist before the first survey, the extension also yields a clean pre-survey share of the decline: 17% in Indonesia over 1985–1993 and 9% in China over 1982–1988. Both are of the same order as the post-1990 survey estimates for those countries, and both place additional family-size-driven poverty reduction before the window the surveys observe, so the survey-era estimates omit this earlier episode of the same channel.

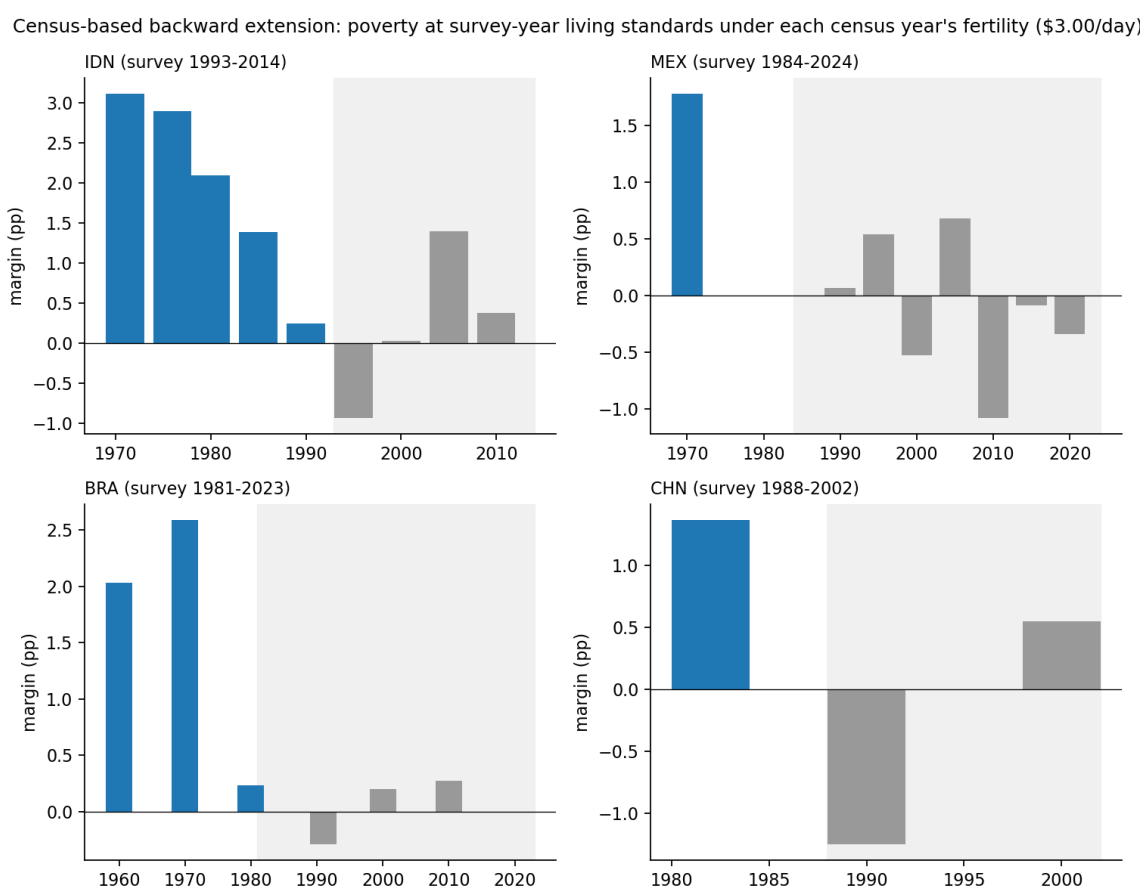


Figure 6: Census-based backward extension of the decomposition for the countries whose census and survey fertility agree at the overlap. The extension reaches 1971 in Indonesia, 1960 in Brazil, 1970 in Mexico, and 1982 in China.

D Robustness: welfare concept, scales, tie-breaking, margins

The estimand holds each household’s measured welfare aggregate fixed and moves only the children distribution. It therefore needs no income-versus-consumption adjustment and no per-child consumption offset: a household that consumes less because it is smaller is a behavioral counterfactual—a different, non-identified quantity—whereas the accounting decomposition asks only how the measured headcount moves when the size denominator changes. For the same reason, chaining a transport across a wave where the welfare concept changes (income to consumption, as in China and Peru) does not bias the decomposition: each wave keeps its own PIP-anchored welfare and only the fertility marginal is swapped.

Table 5 bears this out. No transport leg spans the break—the forward leg uses only the earlier wave’s welfare, the backward leg only the later wave’s—so the concept change never enters the arithmetic. China’s cross-concept leg (income to consumption, 1988–1995) returns a family-size share of 13%, squarely within the range of its within-concept leg (9%) and its overall estimate (11%). The income-only and consumption-only country pools differ (27% versus 12%), but only because the two income-concept countries are slow-growth, late-transition cases—the same growth gradient as in the main results—not because of the welfare concept.

Table 5: The family-size share across the income/consumption welfare-concept break

Series	Window	Margin (pp)	Share (%)
China, income $\rightarrow$ consumption (cross)	1988–1995	2.3	13
China, consumption (within)	1995–2002	1.3	9
Pooled, consumption-only countries	—	—	12
Pooled, income-only (Mexico, Brazil)	—	—	27

Notes: China’s cross-concept leg (income to consumption, 1988–1995) gives a share (13%) in the range of its within-concept leg (9%) and its overall estimate (11%), so crossing the welfare-concept break is not anomalous. The pooled income-vs-consumption gap reflects growth speed—the income-concept countries are slow-growth, late-transition cases—not the concept.

Equivalence scales and poverty lines. Table 6 reports the pooled family-size share under the World Bank’s per-capita metric and three alternative scales, and at the \$4.20 and \$8.30 lines. Because the transport moves one mother’s children at a time, it changes household size by little, so the per-capita and sub-proportional metrics nearly coincide and the share is stable across the higher lines—in contrast to a household-size transport, where the choice of scale matters more. Conditioning on the mother is what makes the result scale-robust.

Tie-breaking and bounds. The unidentified part of the estimate is how ties are broken among mothers with equal children counts. Table 2 reports the share when counterfactual children are assigned rich-first and poor-first; the random headline lies between these welfare-ordered extremes, and pooled over the clean-decline countries the band is 2%–23%. The assumption-free bounds that drop rank preservation entirely span zero—the quantitative statement that rank preservation, supported by the gradient diagnostics, carries the identification.

Table 6: Family-size share under alternative equivalence scales and poverty lines

	Pooled share (%)	Country range (%)
Equivalence scale (\$3.00/day):		
Per capita (World Bank line)	13	7–44
Square root ($\sqrt{n}$)	13	9–39
OECD-modified	13	8–37
Child = 0.5 adult	13	7–37
Poverty line (per capita):		
\$3.00/day (headline)	13	7–44
\$4.20/day	13	6–35
\$8.30/day	14	3–23

Notes: Pooled family-size share of the all-person poverty decline and the country range, decline-weighted over the clean-decline countries (a slightly different base from the people-weighted headline, so levels differ by a point). The mother-level share is stable across equivalence scales — unlike a household-size transport, moving one mother’s children changes household size by little, so the per-capita and sub-proportional metrics nearly coincide. The share is also stable across the \$4.20 and \$8.30 lines.

Extensive versus intensive margins. I also checked whether the effect runs through the extensive margin—a mother’s move across the childless boundary—rather than fewer children among mothers who have any. At the household level, where the split is well defined, the intensive margin (fewer second, third, and higher-order children) accounts for 93% of the effect and the extensive margin only 7%. Poor households sit far from the zero–one boundary; their poverty falls when a family goes from five children to three, not from one to none.

E Related work

The closest prior work and the distinguishing features of this paper are summarized in Table 7. Relative to growth/redistribution accounting (Datt and Ravallion, 1992), this paper transports the size denominator the decomposition conditions on; relative to the economies-of-scale literature (Lanjouw and Ravallion, 1995; Jolliffe and Tetteh-Baah, 2024), it reads a trend rather than a level sensitivity; relative to macro and dependency-ratio accounts (Eastwood and Lipton, 1999; Ravallion, 2016; Wietzke, 2020; Cruz and Ahmed, 2018), it works within household microdata at the level of the mother; and relative to differential-fertility and cohort decompositions (Vogl, 2016; Armentano, Niehaus and Vogl, 2026), it holds household resources fixed and reads exact headcount counterfactuals.

Table 7: Closest papers and how this paper differs

Paper	Question	Data	Method	How this paper differs
Datt and Ravallion (1992)	Decompose a change in poverty	Brazil, India	Growth vs. redistribution on the per-capita distribution	Their split starts after the denominator has changed; I isolate the family-size component inside “growth.”
Lanjouw and Ravallion (1995); Jolliffe and Tetteh-Baah (2024)	Sensitivity of poverty to economies of scale	Cross-country	Level comparisons under alternative scales	I decompose the trend, not the level, of the per-capita denominator.
Eastwood and Lipton (1999); Wietzke (2020); Cruz and Ahmed (2018)	Demographic change and poverty	Cross-country panels, DHS	Macro regressions, dependency ratios	I use a within-household microdata transport, not a macro regression.
Armentano, Niehaus and Vogl (2026)	How poverty fell	5 countries, cohorts	Cohort decomposition	I isolate the denominator component of per-capita welfare via a distribution transport at near-global coverage.
This paper	How much of the \$/day decline is the family-size denominator	22 countries (micro), census extension before surveys	Rank-preserving children-per-mother transport at fixed household consumption	—

Notes: Qualitative comparison of the closest prior work. The distinguishing feature is a within-household, microdata transport that holds household resources fixed and reads exact headcount counterfactuals, rather than a macro regression, a level comparison, or a cohort decomposition.